AAE 53900 Rocket Propulsion
3 credits

1 Course Information/Description

Generalized internal compressible and incompressible flows. Thermochemistry and heat transfer in rocket applications. Advanced topics in solid rocket motor performance and internal ballistics. Liquid rocket engine cycle analysis and turbopump design. Combustion of liquid and solid propellants. Hybrid rockets and thermal-nuclear engines.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:

- Heister S. D., Anderson W. E., Pourpoint T. L., Cassady R. J., Rocket Propulsion, Cambridge University Press - Online ISBN: 9781108381376. [Optional but recommended]
- Sutton G. P. and Biblarz, O. Rocket Propulsion Elements, 7th, 8th or 9th Edition, Wiley & Sons, 2001/2011/2017. [Optional]

3 Assignments

- All homework assignments will be due on Gradescope by the date and time noted on the assignments.
- A tutorial on how to submit assignments to Gradescope can be found here: https://www.purdue.edu/innovativelearning/supporting-instruction/instructional-technology/docs/Vendor_Guide_for_Student_Submissions.pdf
- Late assignments will be not be accepted for any reason although the lowest homework score will be dropped.
- Homework must be neat, legible, and organized. You must understand the work you turn in.
- In general, we will not share solutions of the homework assignments, instead, you should join our office hours to ask for clarifications.
- You can expect five homework assignments each requiring about 1 week to complete and two homework assignments each requiring about 2 weeks to complete this semester with the **nominal** dates listed on our Brightspace page. These nominal dates are subject to change but given to you to help you plan your work.
- Copying someone else's work is not allowed - copied homework will receive a grade of zero and a report will be submitted to the Office of the Dean of Students.

- Any use of Artificial Intelligence or Large Language Models (AI/LLM) to generate parts of a homework or exam solution will be considered a blatant attempt at cheating and will result in a failing grade in the course.

4 Course Learning Objectives

1. Establish solid understanding of rocket fundamentals and thermochemistry
2. Develop capability to study steady, one-dimensional flows applied to the design of rocket engines
3. Apply heat transfer methods to the design of rocket engine cooling systems
4. Develop solid understanding of solid, liquid, and hybrid chemical propulsion systems
5. Develop solid understanding of rocket turbomachinery and/or electric propulsion fundamentals depending on time/interest

5 Evaluation

- Homework (1 week long each)– 25%
- Homework (2 weeks long each)– 15%
- Midterm Exam – 25%
- Final exam – 35%

A grade is assigned at the end of the semester based on a curved scale of all students in class. Those who have demonstrated an excellent understanding of rocket propulsion will be given an A+, A, or A- grade. Those who have demonstrated a minimum acceptable understanding will receive a C+, C, or C- grade with B grades falling in between these extremes. Those who have demonstrated an unacceptable level of understanding will receive a D or F grade.